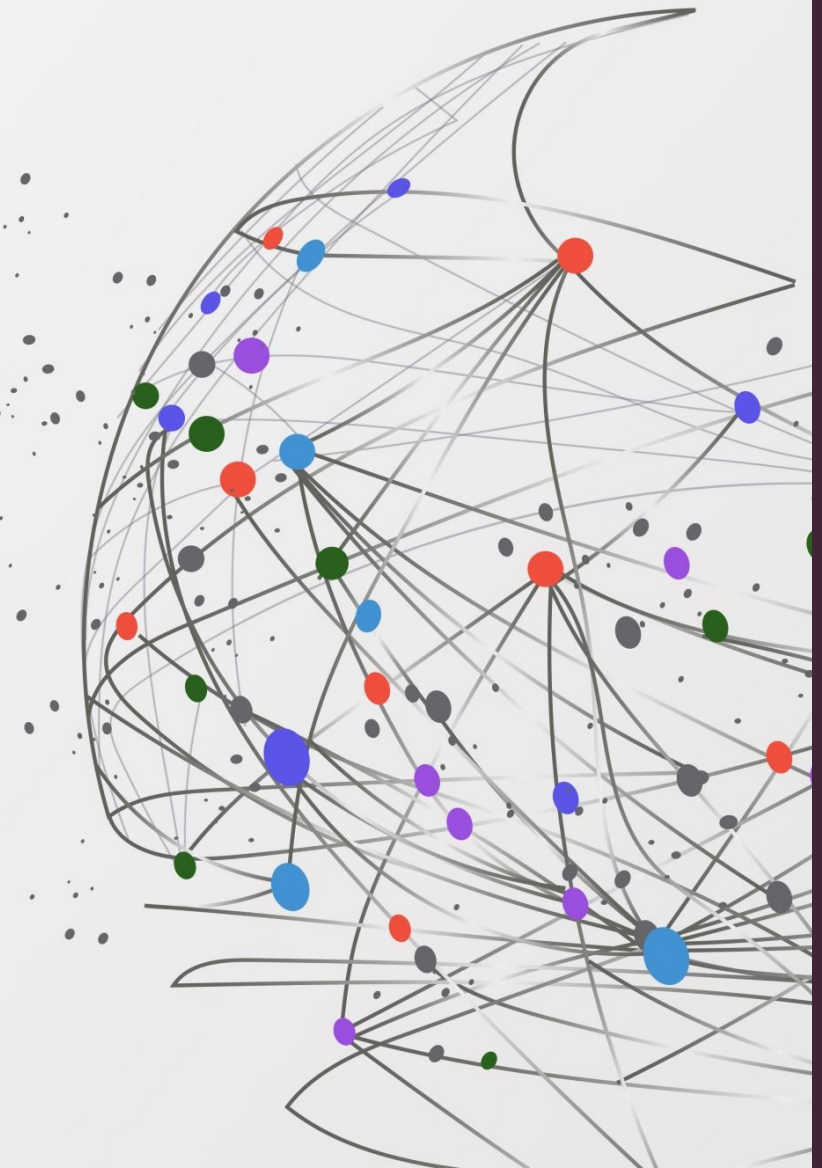


ENSURING SECURE MULTIMEDIA DATA PROCESSING IN IOT DEVICES

Bradley Allen

ABSTRACT

- Introduction/Motivation
- Overview of each paper
- Conclusion and Future Work
- References

INTRODUCTION

- Internet of things (IoT) is a very fast-growing concept.
- Security, cybersecurity, and privacy.
- IoT consists of many multimedia devices.
- IoT search engines
- Cyber attacks using IoT technology.
- Development of Internet of Things (IoT).
- Latest use of IoT multimedia technologies.

WHAT DOES MEAN SEARCH ENGINE FOR IOT OR IOT SEARCH ENGINE

First paper

SEARCH ENGINES FOR MACHINE LEARNING

- Data collection of multimedia through the internet.
- Personal information taken and stored.
- IP addresses, Ports, Locations, Hostname, devices, and coordinates.
- Pool of information about users.

PROBLEMS

- Security and Privacy issues.
- Database security precautions.
- User information at stake.
- Break can lead to major damage.

	Shodan	Censys	Topical Search Engine for IOT
Founding date	2009	2015	2014
Website	www.shodan.io	www.Censys.io	-
Founders	John Matherly	Zakir Durumeric	Faqeeh M. et al.
IP Address	IPv4 & IPv6	IPv4	-
Data collection	Port scanner : SYN Scan	Port scanner : ZMap (+ZGrab)	crawler
Data stores in Database	device summary (IP address, Location, latitude / longitude	very complete data about IOT devices	URLS
	coordinates) and available services		• •
Access	web interface, API	web interface, API or texts to download	web interface Search •
Filters	IP, Port, Location, version, hostname…	Keyword, IP, protocol used, certificate....	Keyword • Device

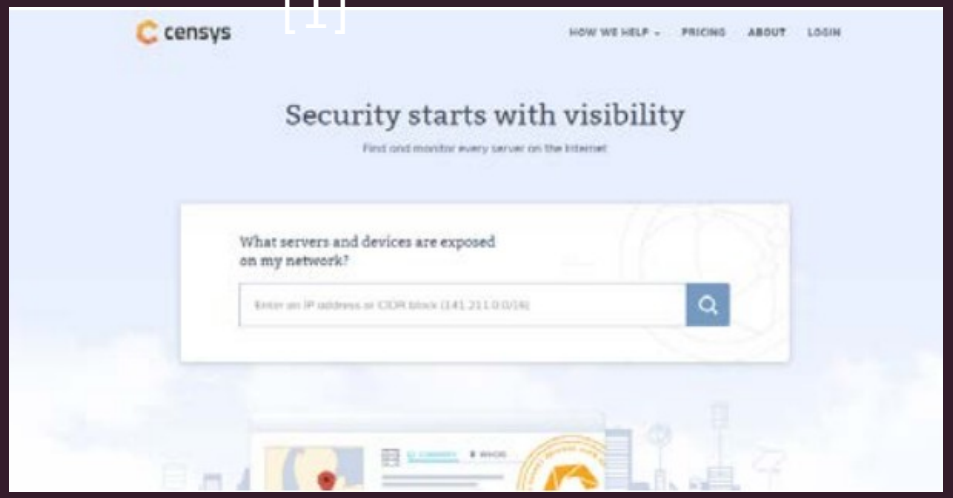
Figure 6 from [1]

Search Engines for IoT connections

Security or Spying?

Example an IoT search Engine

Figure 5 from [1]



https://search.censys.io/

FIRST PAPER COVERS THE SEARCH OF ENGINES OF IOT

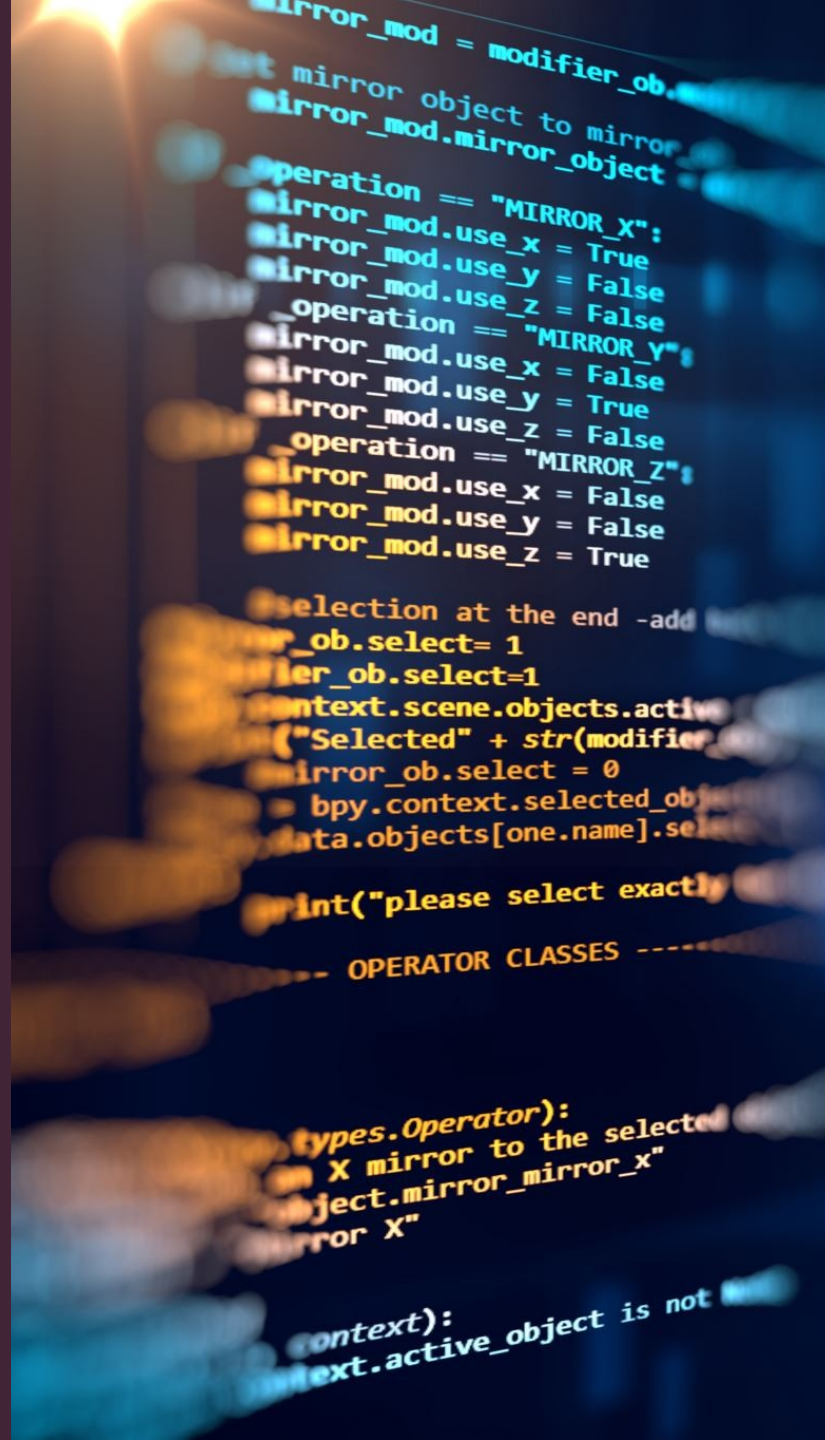
- Such as the use of search engines.
- How data is being searched
- How data is being stored.
- All regarding multimedia devices and processes.

TOWARDS DESIGNING OPEN AND SECURE IOT SYSTEMS: INSIGHTS FOR PRACTITIONERS

Second paper

DEVELOPMENT OF IOT SYSTEM PROBLEMS

- Security not prioritized as part of development cycle.
- Interviews with many types of roles in IoT domains lack security awareness.
- Challenges developing security and privacy default.
- Lack of resources.
- Large area for IoT too big to focus on all.
- Awareness and Mindset problems.



SOLUTIONS

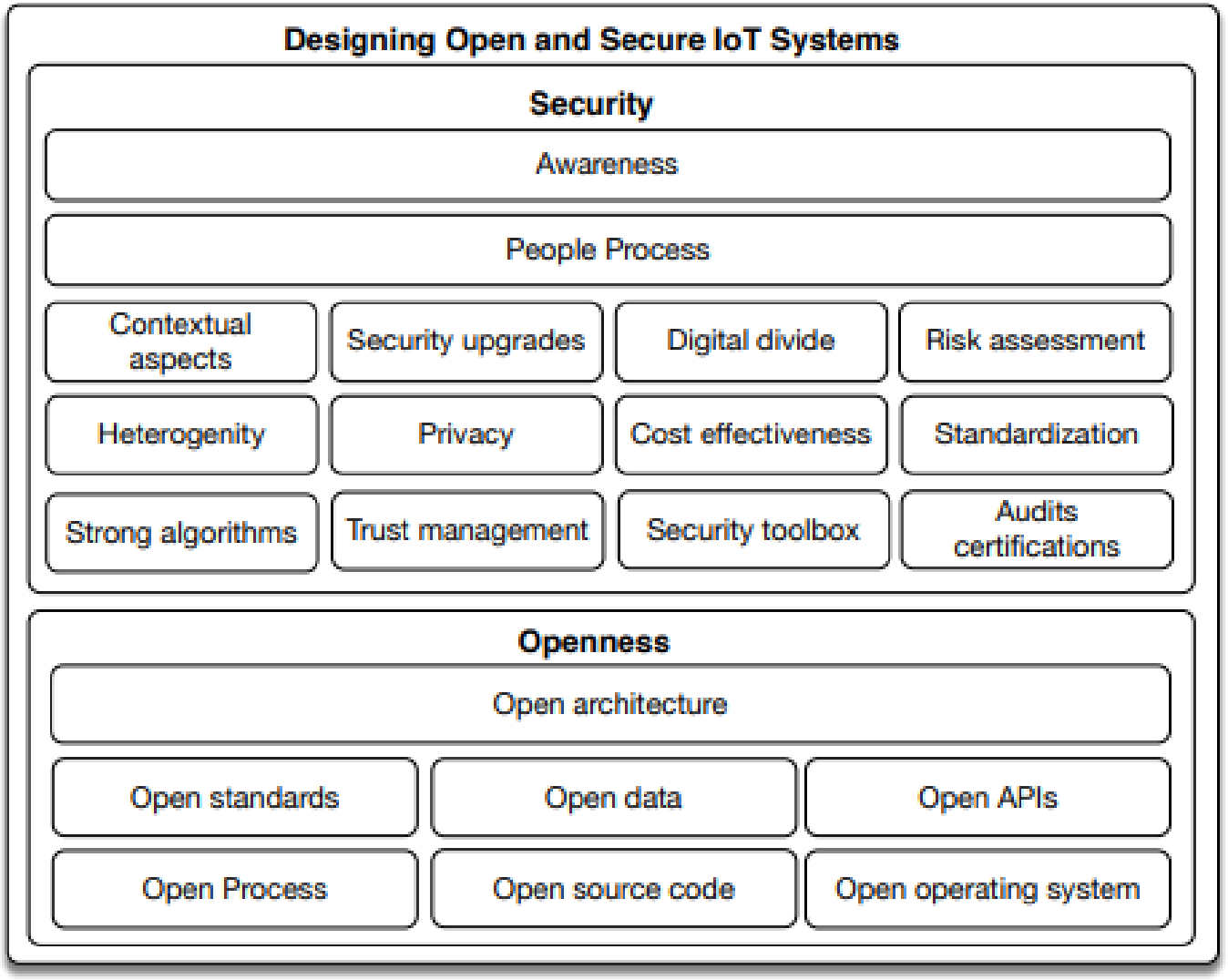


Figure 1. Open and Secure IoT design model conceptualized.

- Figure 1 from [2]

SECOND PAPER COVERS DEVELOPMENT OF IOT SYSTEMS AND LACK OF SECURITY WITHIN IT.

- To ensure secure multimedia processing with IoT devices security must be developed with it.

Ensuring Secure Multimedia Data Processing in IoT Devices

IOT LOTTO: UTILIZING IOT DEVICES IN BRUTE-FORCE ATTACKS

Third paper

ATTACKS USING IOT SYSTEMS

- Brute force attacks
- Network “man in the middle attacks”
- Using information from multimedia devices.
- Device Hijacking

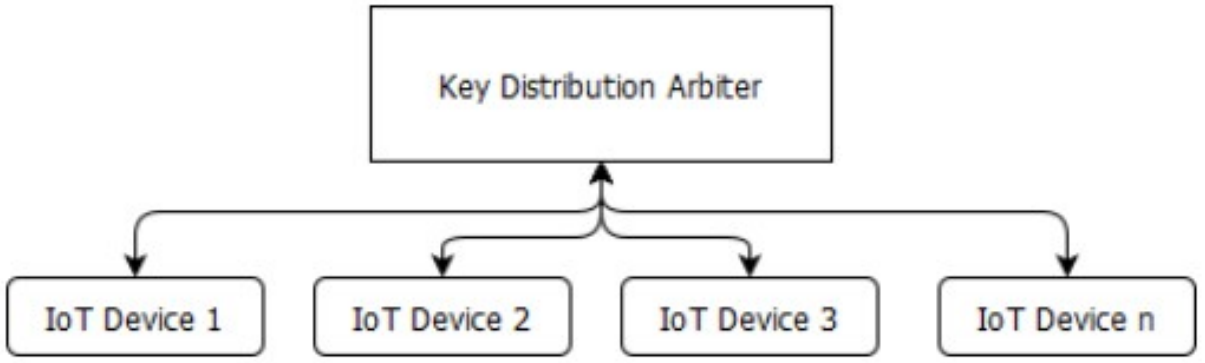


Figure 1: Overview of IoT Lotto

- Figure 1 from [3]

IOT CAN BE DANGEROUS

- The paper explains how IoT devices such as TVs can be used to decrypt keys.
- Imagine what else these devices can be used for.
- Security should be topmost importance.

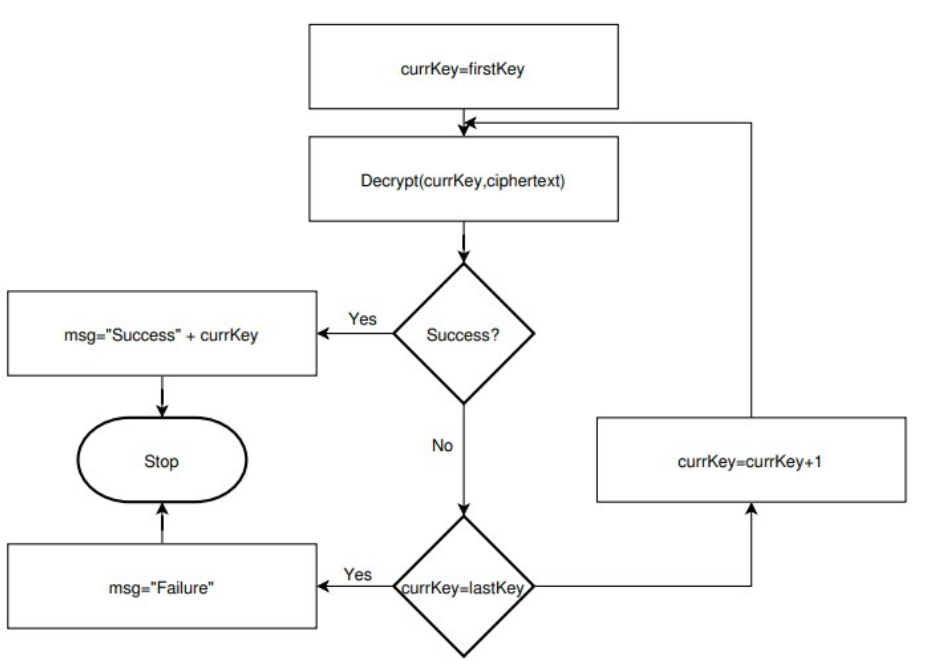
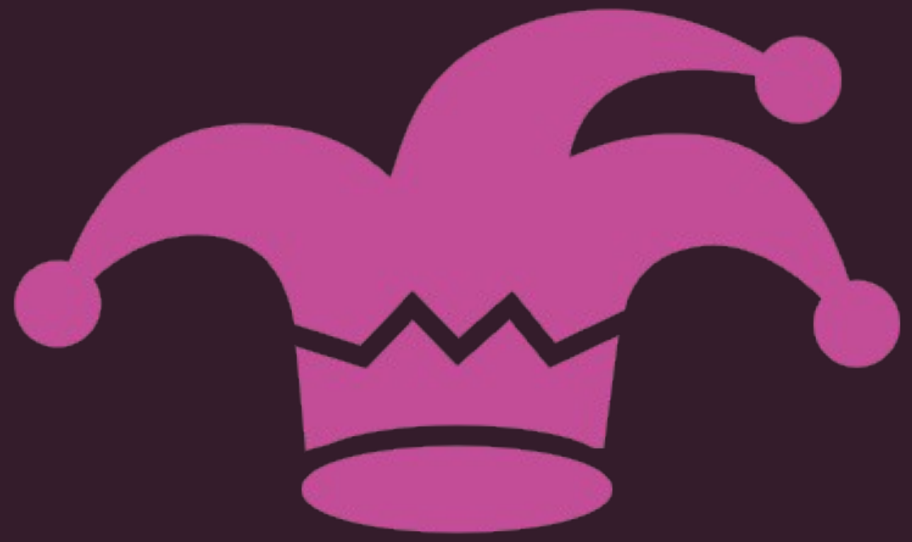


Figure 3: Flowchart of key generation and decryption at the IoT device

- Figure 3 from [3]

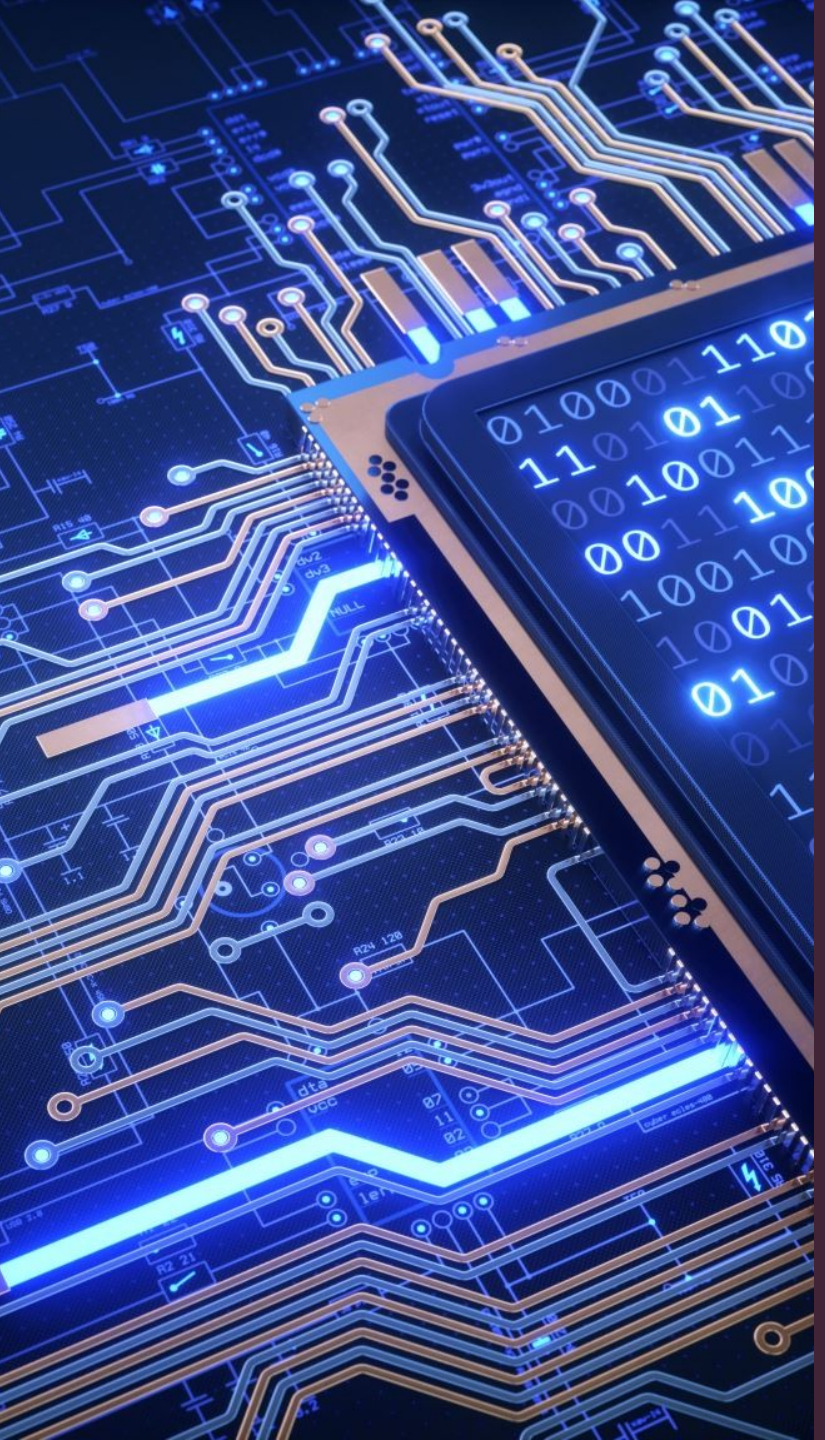
THIRD PAPER GOES OVER SECURITY THREATS

- Threats for these multimedia device, involve using its processing power and data to cater to vulnerability.
- These threats can lead to crime, and cause problems for the users.



ENHANCING PUBLIC TRANSIT ACCESSIBILITY FOR THE VISUALLY IMPAIRED USING IOT AND OPEN DATA INFRASTRUCTURES

Fourth paper



ADVANCEMENTS OF IOT MULTIMEDIA DEVICES

- Assistive device for those with disability.
- Allowing data and information to be transferred to and from multimedia devices
- IoT used for improving lives.

SECURITY THREATS THAT COME WITH ADVANCEMENTS

- Encryption/Decryption
- Device Hijacking
- Security Development?
- Misinformation
- Data leaks for users.
- True/False data.

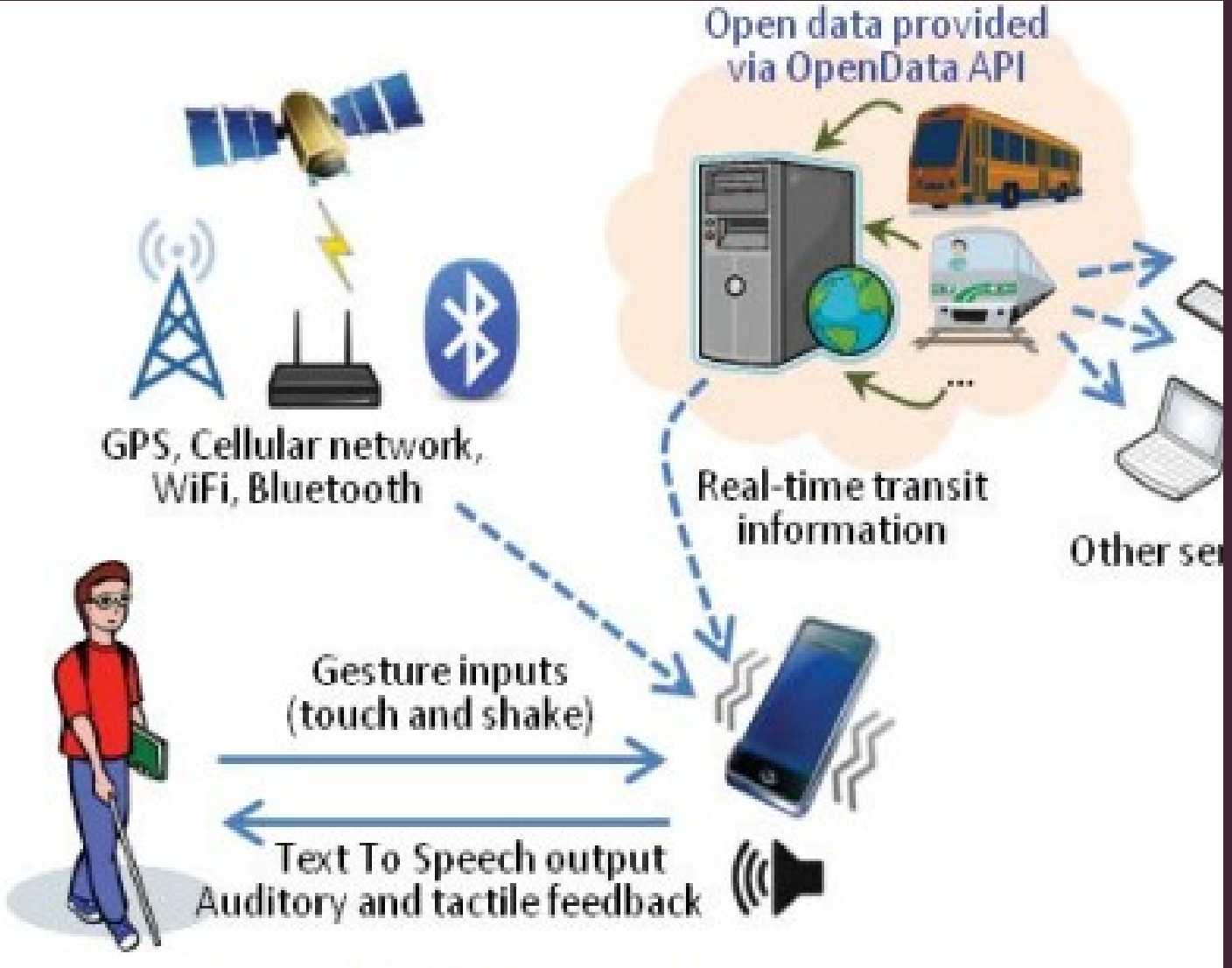
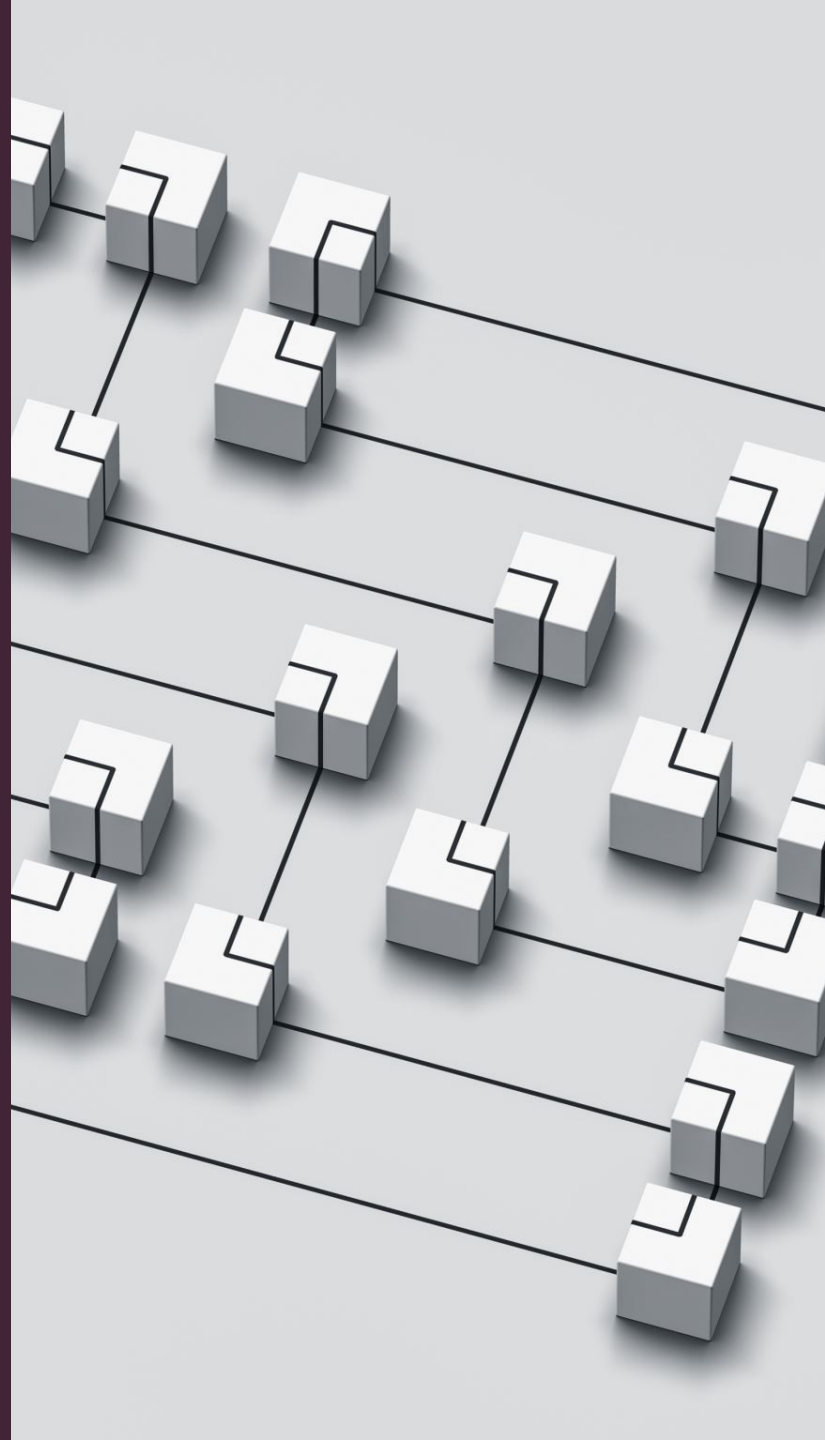


Figure 1. TalkingTransit system architecture.

FOURTH PAPER CONSISTS OF DEVELOPING AN AID FOR BLIND IN PUBLIC TRANSIT

- This improvement and development of the system is amazing for the world.
- Lacking in security features and privacy methodology's
- Good example of a IoT system that does not have parallel development in security as it does in other areas.



CONCLUSION

All of the papers have a common feature:

They each have a major improvements in IOT in some way

Each have security features to them whether it be good or bad.

IoT for multimedia processing and devices will continue to grow in the future.

The development of ensuring security with the devices need and will grow also.

FUTURE WORK

I believe that even though the internet is not a new technology, the use of the internet and the things that are contained will be new.

Improvements with the multimedia devices connected to the internet will continue to grow.

The entire of the internet and technology will continue to grow at a fast pace.

Security will undoubtedly grow as the threats and technology do.

THANK YOU

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